Intermediate Microeconomics

Week 3 · Day 2 — Introduction to Game Theory

Summer School

From "mutual best replies" in duopoly to a general language for strategic choice.

From yesterday's duopoly to today's toolkit

Every duopoly outcome yesterday was a *resting point*: a pair of choices where **neither firm wished to move, given what the other was doing**.

- Cournot: $q_1 = 30$ was firm 1's best reply to $q_2 = 30$, and vice versa.
- Bertrand: $P = 30$ — no one gains by changing price.
- Stackelberg: the follower best-replied; the leader had already optimised against that.

Today

We give this idea its proper name — **Nash equilibrium** — and build a toolkit that works far beyond duopoly: prisoners, coordination, bluffing, and moves made in sequence.

Learning goals — Day 2

1. Describe a game by its *players, strategies, and payoffs*, and read a payoff matrix.
2. Define a **best response**, a **dominant strategy**, and a **Nash equilibrium**.
3. Solve the **Prisoner's Dilemma** and see it as the logic behind a price war.
4. Handle games with *several* equilibria (**coordination**) and with *none in pure strategies* (**zero-sum**).
5. Compute a **mixed-strategy** Nash equilibrium.
6. Draw a game in **extensive form** (a tree) and solve it by **backward induction**.

What is a game?

A (non-cooperative) game has three ingredients:

Ingredients

1. **Players** — who decides (firm 1, firm 2; Ann, Bob).
2. **Strategies** — the actions each can take (a price; a quantity; "cooperate" / "defect").
3. **Payoffs** — the number each player earns for every *combination* of strategies.

A two-player game with finite strategies is written as a **payoff matrix**. Convention: the *row* player's payoff is listed first, the *column* player's second.

	Left	Right
Up	(a_1, a_2)	(b_1, b_2)
Down	(c_1, c_2)	(d_1, d_2)

Best response, dominant strategy, Nash equilibrium

Three definitions

- A strategy is a **best response** to the rival's choice if it gives the highest payoff *against that choice.*
- A **dominant strategy** is a best response to *every* choice the rival could make — always optimal.
- A **Nash equilibrium (NE)** is a strategy profile in which *every* player is simultaneously best-responding: no one can gain by unilaterally deviating.

This is exactly yesterday's "resting point"

In Cournot, $(q_1, q_2) = (30, 30)$ is a Nash equilibrium: each output is a best response to the other. The reaction curves crossed *because* that is where both best-response conditions hold at once.

The Prisoner's Dilemma — a price war

Two firms each choose a **High** price (tacitly collude) or a **Low** price (undercut). Profits ($000s), row = firm 1:

	Firm 2: High	Firm 2: Low
Firm 1: High	(100, 100)	(20, 140)
Firm 1: Low	(140, 20)	(60, 60)

Find firm 1's best response (read down each column)

- If firm 2 plays High: $140 > 100$, so firm 1 plays **Low**.
- If firm 2 plays Low: $60 > 20$, so firm 1 plays **Low**.

Dominant strategy

Low is a best response to *both* of firm 2's choices — it is a *dominant strategy* for firm 1, and by symmetry for firm 2.

The dilemma: individually rational, jointly bad

Both firms play their dominant strategy **Low**, so the unique Nash equilibrium is

$$(\text{Low}, \text{Low}) = (60, 60).$$

The tension

Both would be better off at $(\text{High}, \text{High}) = (100, 100)$! But that is *not* an equilibrium: each firm is tempted to undercut $(140 > 100)$. Self-interest destroys the collusive profit.

Why this is yesterday's Bertrand

Price-cutting is a dominant strategy, so price is driven down and profits collapse — exactly the Bertrand logic. The Prisoner's Dilemma is the *strategic skeleton* of price competition.

The forward question

If the firms met *again and again*, could the threat of future price wars sustain $(100, 100)$? That is **Day 3**.

Coordination — when there is more than one equilibrium

Ann and Bob want to spend the evening *together*, but Ann prefers the Opera and Bob the Football (the "Battle of the Sexes"). Row = Ann:

	Bob: Opera	Bob: Football
Ann: Opera	(2, 1)	(0, 0)
Ann: Football	(0, 0)	(1, 2)

Look for mutual best responses

- $(\text{Opera}, \text{Opera})$: Ann's $2 > 0$ ✓, Bob's $1 > 0$ ✓— a Nash equilibrium.
- $(\text{Football}, \text{Football})$: both > 0 against the alternative ✓— *also* a Nash equilibrium.
- The mismatched cells give $(0, 0)$: someone always wants to switch.

Two equilibria

Being together matters more than which event — but which one? Theory alone does not say.

What coordination games teach

- Nash equilibrium need *not* be unique — theory alone does not always pick one outcome.
- Real coordination is resolved by **focal points**, **conventions**, communication, or who moves first.
- Examples: which technology standard an industry adopts; driving on the left vs. right; two firms choosing compatible vs. incompatible formats.

Prisoner's Dilemma vs. Coordination

- *Prisoner's Dilemma*: one equilibrium, and it is the *bad* one (conflict of interest).
- *Coordination*: several equilibria, all "good" — the problem is *selecting* one.

There is also a *third* equilibrium here, in which each player randomises. To find it we need mixed strategies — but first, a game with *no* pure equilibrium at all.

Zero-sum games — pure conflict

In a **zero-sum** game one player's gain is exactly the other's loss: the payoffs in every cell sum to zero. **Matching Pennies**: each shows Heads or Tails. Player 1 wins if they *match*, Player 2 if they *differ*.

	P2: Heads	P2: Tails
P1: Heads	$(+1, -1)$	$(-1, +1)$
P1: Tails	$(-1, +1)$	$(+1, -1)$

Is there a pure-strategy equilibrium?

Check each cell: in (H, H) player 2 wants to switch to Tails; in (H, T) player 1 wants to switch; … **every** cell has someone who wants to deviate.

No pure-strategy Nash equilibrium exists

If you always play Heads, your opponent exploits you. The only way to be unpredictable is to *randomise*.

Mixed strategies — randomise to stay unpredictable

A **mixed strategy** plays each action with some probability. Let player 1 play Heads with probability p, player 2 play Heads with probability q.

The indifference principle

In a mixed equilibrium, each player randomises *only if indifferent* between their actions. So player 1 chooses p to make **player 2** indifferent, and player 2 chooses q to make **player 1** indifferent.

Make player 1 indifferent (choose q)

$$\text{from Heads}: \; q(+1) + (1-q)(-1) = 2q - 1, \qquad \text{from Tails}: \; q(-1) + (1-q)(+1) = 1 - 2q.$$

Solve

$2q - 1 = 1 - 2q \Rightarrow 4q = 2 \Rightarrow q = \frac{1}{2}$; by the symmetric calculation for player 2, $p = \frac{1}{2}$.

The mixed equilibrium — reading it

Each player plays Heads and Tails with probability $\frac{1}{2}$.

- Expected payoff to each: 0 — a fair game.
- Neither can be exploited: any deviation is met by an equally-good reply.

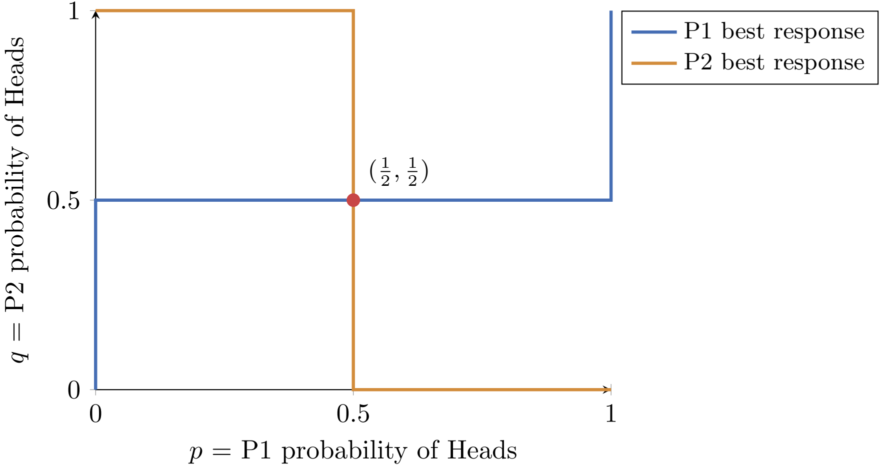
Best-response "step" curves (drawn next)

- If $q > \frac{1}{2}$ (rival mostly Heads), P1 wants to match: $p = 1$.
- If $q < \frac{1}{2}$, P1 plays Tails: $p = 0$.
- At $q = \frac{1}{2}$, P1 is indifferent — *any* p.

The equilibrium

The two best-response curves cross *only* at $(p, q) = (\frac{1}{2}, \frac{1}{2})$.

The mixed equilibrium — the picture



The step-shaped best responses meet at a single point: each mixing 50/50 is the unique equilibrium.

Mixing in the coordination game (the third equilibrium)

Back to Ann and Bob. Let Ann play Opera with prob. p, Bob play Opera with prob. q. **Make Bob indifferent (choose Ann's p)**

Bob's payoff to Opera $= p \cdot 1 + (1-p) \cdot 0 = p$; to Football $= p \cdot 0 + (1-p) \cdot 2 = 2(1-p)$.

$$p = 2(1-p) \;\Rightarrow\; 3p = 2 \;\Rightarrow\; p = \tfrac{2}{3}.$$

Make Ann indifferent (choose Bob's q)

Ann's payoff to Opera $= q \cdot 2$; to Football $= (1-q) \cdot 1$.

$$2q = 1 - q \;\Rightarrow\; 3q = 1 \;\Rightarrow\; q = \tfrac{1}{3}.$$

Result

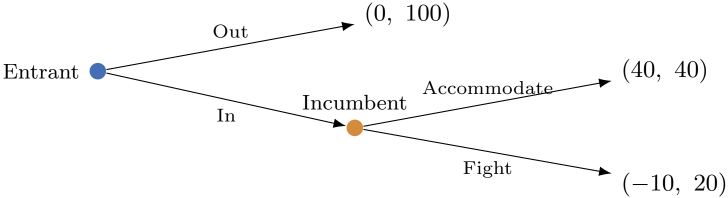
A third (mixed) equilibrium: Ann plays Opera $\frac{2}{3}$ of the time, Bob plays Opera $\frac{1}{3}$ of the time. They *miscoordinate* often — mixed equilibria can be inefficient.

When players move in sequence: the extensive form

Not all games are simultaneous. When one player moves *first* and the other *sees* it, we draw a **game tree** (extensive form). Yesterday's Stackelberg leader–follower was exactly this.

An entry game

An **Entrant** decides whether to enter a market. If it enters, the **Incumbent** decides whether to *Fight* a price war or *Accommodate*. Payoffs = (Entrant, Incumbent):



Backward induction — solve the tree from the end

Step 1 — the last mover (Incumbent)

If the entrant has entered, the incumbent compares

$$\text{Accommodate} : 40 \quad > \quad \text{Fight} : 20 \quad \Rightarrow \quad \text{the incumbent } \textbf{accommodates}.$$

Step 2 — the first mover (Entrant)

Anticipating accommodation, the entrant compares

$$\text{In} : 40 \quad > \quad \text{Out} : 0 \quad \Rightarrow \quad \text{the entrant } \textbf{enters}.$$

Solution

(In, Accommodate), with payoffs $(40, 40)$.

Credible and non-credible threats

Non-credible threats

The incumbent might *threaten*: “enter and I'll fight!” But the threat is **not credible**: once entry has happened, fighting (20) is worse than accommodating (40). A rational entrant ignores it and enters.

Link to Day 1

This is why the Stackelberg *leader* gained: by committing to a large quantity *first*, it changed what was credible afterwards. Commitment is the currency of sequential games.

The general lesson

Backward induction rules out equilibria that rest on threats a player would not actually carry out. To deter entry for real, the incumbent must change the payoffs *before* the entrant moves (e.g. build capacity).

Simultaneous vs. sequential — the order of moves matters

Same players, same payoffs, different timing

- **Simultaneous** (normal form, payoff matrix): no one sees the other's move — Cournot, Prisoner's Dilemma, Matching Pennies.
- **Sequential** (extensive form, tree): later movers observe earlier ones — Stackelberg, the entry game. Solve by *backward induction*.

- Moving first can be an **advantage** (Stackelberg leader: commit to a large output).
- Seeing first can also help — it depends on the game.
- Backward induction rules out equilibria that rest on *non-credible threats*.

Day 2 checkpoint

You can now

- write a game as players/strategies/payoffs and find best responses;
- locate **Nash equilibria** — and recognise the **Prisoner's Dilemma** in a price war;
- handle **multiple** equilibria (coordination) and **no pure** equilibrium (zero-sum);
- compute a **mixed-strategy** equilibrium via the indifference principle;
- solve a **sequential** game by backward induction and spot non-credible threats.

Tomorrow

The Prisoner's Dilemma said cooperation collapses in one shot. But firms meet *repeatedly*. **Day 3**: when does the shadow of the future let rivals sustain the good outcome?

In-class exercises — Day 2

Time for the in-class exercises.

- **Exercise 1** — dominance and Nash equilibrium in 2×2 games; identify the game type.
- **Exercise 2** — a mixed-strategy equilibrium and a game tree solved by backward induction.

Work in pairs; for each game be ready to say how many equilibria it has and of what type.